COMPUTER NETWORKING & CISCO PACKET TRACER

Complete MCQ Examination Study Notes

Prepared from the two uploaded syllabus images

Examination format: 60 MCQs | 120 minutes | Average time: 2 minutes per question

Includes networking fundamentals, OSI and TCP/IP, protocols, subnetting, Cisco IOS, routing, switching, security, NAT, troubleshooting and practical configuration.

Study approach: understand the concept → learn the comparison → practise the command or calculation → solve the MCQs.

How to use these notes

These notes follow the syllabus in the uploaded images. Part A covers networking theory. Part B covers Cisco Packet Tracer and administration labs. Commands are written in Cisco IOS-style syntax; exact command availability can vary by router, switch model, IOS version and Packet Tracer device.

- First read the concept and diagram; do not memorise commands without knowing what they change.
- For subnetting, practise the worked examples with a blank sheet and verify network, usable-host and broadcast addresses.
- For routing and switching, learn both the purpose of each protocol and the configuration/verification commands.
- In MCQs, watch for words such as always, only, best, same network, next hop, local, remote, reliable and connectionless.
- Use the final revision sheet immediately before the examination.

Examination time strategy

Stage	Suggested time	Approach
First pass	70–80 min	Answer direct definition, port, layer, device and command questions.
Second pass	25–30 min	Solve subnetting, routing, ACL and scenario questions.
Review	10–15 min	Recheck marked questions, negative wording and calculations.
Total	120 min	60 questions; average 2 minutes each.

EXAM ALERT: Do not spend five minutes on one difficult question during the first pass. Mark it, move on and return later.

PART A — COMPUTER NETWORKING FUNDAMENTALS

1. Living in a Network-Centric World

1.1 What is a computer network?

A computer network is a group of two or more computing devices connected through wired or wireless communication links so that they can exchange data and share resources. Devices may include computers, servers, phones, printers, cameras, switches and routers. A network needs endpoints, a medium, addressing and agreed communication rules (protocols).

1.2 How networks support daily life

- **Communication:** email, instant messaging, voice calls, video meetings and social media.
- **Resource sharing:** shared printers, storage, applications and internet connections.
- **Business:** cloud applications, online banking, e-commerce, inventory and remote work.
- **Education:** virtual classrooms, learning-management systems and digital libraries.
- **Public services:** telemedicine, online government services, transport systems and emergency communication.
- **Entertainment:** streaming, multiplayer gaming and content delivery networks.

1.3 Network characteristics

Characteristic	Meaning	Example / exam clue
Fault tolerance	Network continues operating when a component/link fails, often through redundancy.	Alternate link or redundant router.
Scalability	Can grow without unacceptable performance loss or redesign.	Add users, switches, sites or bandwidth.
Quality of Service (QoS)	Prioritises selected traffic to meet delay, jitter or loss requirements.	Voice/video prioritised over bulk downloads.
Security	Protects confidentiality, integrity and availability.	Authentication, encryption, firewall, access control.
Availability	How consistently the network is usable.	Redundant power, links and devices.

1.4 Network devices

Device	Primary function	Important distinction
NIC (Network Interface Card)	Connects an endpoint to a network; provides a Layer 2 interface and usually a MAC address.	A NIC is installed in/attached to a host.
Hub	Multiport Layer 1 repeater; repeats incoming bits to all other ports.	No MAC learning; one collision domain.
Repeater	Regenerates/retimes a signal to extend a physical link.	Layer 1; does not route or filter frames.
Switch	Forwards Ethernet frames using a MAC address table.	Typically Layer 2; each port is a separate collision domain.
Router	Forwards IP packets between different networks using a routing table.	Layer 3; separates broadcast domains by default.
Firewall	Enforces traffic policy between security zones or hosts.	Can filter by address, port, state or application.
Load balancer (LB)	Distributes client requests across multiple servers.	Improves availability and capacity; may operate at L4 or L7.
Modem	Modulates/demodulates signals to connect over a provider access technology.	The word comes from modulator-demodulator.

1.5 Network architecture characteristics

A network architecture describes how devices, services, protocols and links are organised. A **fault-tolerant** design uses redundancy; a **scalable** design supports growth; a **QoS-aware** design manages traffic classes; and a **secure** design applies layered controls. Modern networks commonly use a mix of client-server and peer-to-peer communication.

1.6 Network types

Type	Coverage / purpose	Example
PAN	Personal area; a few metres around a person.	Bluetooth headset and phone.
LAN	Limited area such as room, office or building.	Office Ethernet.
WLAN	LAN using wireless radio, commonly Wi-Fi.	Home Wi-Fi.
CAN	Campus area connecting multiple buildings.	University network.
MAN	Metropolitan/city-scale network.	City-wide provider network.
WAN	Large geographic area; connects distant LANs.	Enterprise branches connected through provider links.

EXAM ALERT: Internet is a global network of interconnected networks. It is not the same thing as the Web; the Web is one service that runs over the Internet.

1.7 Network topologies

Topology	Structure	Strengths	Limitations
Bus	All devices share one backbone cable.	Simple, low cable use in legacy networks.	Backbone failure affects all; collisions; difficult fault isolation.
Star	Each device connects to a central switch/hub.	Easy to manage and expand; one access-link failure affects one device.	Central device failure affects the segment.
Ring	Each device connects to two neighbours in a loop.	Predictable access in some ring technologies.	Break can disrupt ring unless protected.
Mesh	Devices have multiple interconnections.	High redundancy and resilience.	Cost and complexity increase.
Tree / hierarchical	Star networks arranged in layers.	Scalable, structured campus design.	Failure in an upper-layer device can affect many downstream devices.
Hybrid	Combination of topologies.	Flexible design.	More planning and troubleshooting complexity.

Physical topology describes actual cabling/device connections. **Logical topology** describes how data flows. They may differ; for example, switched Ethernet is physically star-shaped.

Quick revision and MCQ traps

- A hub and repeater operate at the Physical Layer; a conventional switch at Data Link; a router at Network.
- A switch learns source MAC addresses; it does not learn IP routes.
- A router separates Layer 2 broadcast domains.
- QoS is about traffic treatment, not simply increasing bandwidth.
- A mesh offers redundancy but costs more.

Common traps: Hub does not create separate collision domains per port. • A switch does not normally forward a Layer 2 broadcast to another VLAN. • LAN describes scope; Ethernet describes a common LAN technology.

Q1. Which device regenerates a signal without examining IP addresses?

A. Router **B.** Repeater **C.** Firewall **D.** Layer 3 switch

Answer: B. A repeater works at Layer 1 and regenerates physical signals.

Q2. Which characteristic allows a network to grow as users and devices increase?

A. Scalability **B.** Encapsulation **C.** Jitter **D.** Modulation

Answer: A. Scalability is the ability to expand while maintaining acceptable operation.

Q3. Which device learns MAC addresses and builds a forwarding table?

A. Hub **B.** Repeater **C.** Switch **D.** Modem

Answer: C. A switch learns source MAC addresses from received Ethernet frames.

Q4. A router normally separates which type of domains?

A. Collision only **B.** Broadcast **C.** Optical **D.** Application

Answer: B. Routers do not forward Layer 2 broadcasts by default, so each routed interface/VLAN is a separate broadcast domain.

Q5. Which topology provides the greatest number of alternate paths when fully meshed?

A. Bus **B.** Star **C.** Full mesh **D.** Single ring

Answer: C. A full mesh has a direct link between every pair of nodes, providing high redundancy.

2. Communication over the Network

2.1 Basic communication components

A communication system has a **sender**, **receiver**, **message**, **medium** and **protocol**. The sender encodes data into signals; the medium carries the signals; the receiver decodes them. Protocols define syntax (format), semantics (meaning) and timing (when/how fast).

2.2 Network communication types

Type	Meaning	Example
Unicast	One sender to one intended receiver.	PC sends a web request to one server.
Broadcast	One sender to all devices in the local broadcast domain.	ARP request in IPv4 Ethernet LAN.
Multicast	One sender to a subscribed group of receivers.	IPTV or group streaming.
Anycast	Same service address announced from multiple locations; routing selects a suitable instance.	Distributed DNS service.

2.3 Host-to-host communication model

Communication is end-to-end: an application creates data, transport provides process-to-process delivery, the network layer provides logical addressing and routing, the data-link layer delivers a frame over one local link, and the physical layer transmits bits. At each router, the incoming frame is removed and a new frame is built for the next link; the IP packet is forwarded toward its destination.

Host A → Switch → Router(s) → Switch → Host B

2.4 OSI reference model

Layer	Name	Main responsibility	PDU / examples
7	Application	Network services used by applications.	Data; HTTP, DNS, SMTP, FTP
6	Presentation	Data representation, translation, compression, encryption concepts.	Data; formats, TLS-related presentation functions conceptually
5	Session	Establishes, manages and terminates dialogue/sessions.	Data; session control
4	Transport	End-to-end process delivery, segmentation, reliability, flow control, ports.	Segment (TCP) / datagram (UDP)
3	Network	Logical addressing and routing between networks.	Packet; IPv4, IPv6, ICMP
2	Data Link	Framing, MAC addressing, media access and link-level error detection.	Frame; Ethernet, PPP, 802.11 MAC
1	Physical	Signals, bits, connectors, media and transmission characteristics.	Bits; copper, fibre, radio

EXAM ALERT: Mnemonic from Layer 7 to 1: All People Seem To Need Data Processing. From Layer 1 to 7: Please Do Not Throw Sausage Pizza Away.

2.5 TCP/IP model and OSI mapping

TCP/IP model (4 layers)	Common responsibilities	OSI layers mapped
Application	Application protocols and data representation/session functions.	OSI 7, 6, 5
Transport	Process-to-process communication.	OSI 4
Internet	Logical addressing and routing.	OSI 3
Network Access / Link	Framing and physical transmission.	OSI 2, 1

Some textbooks show a five-layer Internet model by separating Data Link and Physical from Network Access. Both four-layer and five-layer depictions are used; know which model the question names.

2.6 Encapsulation and de-encapsulation

As data moves down the sender's protocol stack, each layer adds its own control information (headers, and sometimes a trailer). This is **encapsulation**. At the receiver, each corresponding layer interprets and removes its information; this is **de-encapsulation**.

Layer	Typical data unit	Address / information added
Application	Data/message	Application semantics
Transport	TCP segment / UDP datagram	Source and destination port
Network	IP packet	Source and destination IP address
Data Link	Frame	Source and destination MAC address; FCS trailer in Ethernet
Physical	Bits	Signals on the medium

2.7 Peer-to-peer communication

At a given layer, peer entities on two systems logically communicate using that layer's protocol. For example, TCP on one host communicates with TCP on another host, although the actual bits travel down and up the local stacks and through intermediate devices. Peer-to-peer (P2P) can also describe an application architecture in which each node can act as both client and server; do not confuse these two uses.

Quick revision and MCQ traps

- OSI has seven layers; the common TCP/IP model has four layers.
- PDU sequence down the stack: data → segment/datagram → packet → frame → bits.
- Ports identify application processes; IP addresses identify logical interfaces/hosts; MAC addresses identify Layer 2 interfaces on a local link.
- Routers replace Layer 2 framing at each hop; end-to-end IP addressing normally remains, except where NAT changes it.

Common traps: MAC address is not used for end-to-end routing across the Internet. • Encapsulation adds headers; de-encapsulation processes/removes them. • Presentation and Session are distinct OSI layers even when TCP/IP groups their functions.

Q1. Which OSI layer is responsible for logical addressing and routing?

A. Data Link **B.** Network **C.** Transport **D.** Session

Answer: B. Layer 3 handles IP addressing and routing.

Q2. What is the PDU at the Data Link Layer?

A. Bit **B.** Frame **C.** Packet **D.** Segment

Answer: B. Layer 2 encapsulates a network-layer packet into a frame.

Q3. Which information is used by TCP to identify the destination application process?

A. MAC address **B.** Port number **C.** SSID **D.** FCS

Answer: B. Transport-layer port numbers identify application endpoints.

Q4. Which TCP/IP layer combines OSI Application, Presentation and Session?

A. Internet **B.** Transport **C.** Application **D.** Network Access

Answer: C. The TCP/IP Application layer groups those upper OSI functions.

Q5. During forwarding through a router, which item is normally rewritten for the next Ethernet hop?

A. Destination IP always **B.** Ethernet source/destination MAC addresses **C.** TCP destination port **D.** Application payload

Answer: B. A router builds a new Layer 2 frame for the outgoing link.

3. Protocols and Their Functionality

3.1 What is a protocol?

A protocol is a defined set of rules governing communication. It specifies message format, addressing, sequencing, error handling, timing and actions on receiving messages. Open standards enable devices from different vendors to interoperate.

3.2 Human interface and data networks

Humans create information through applications and user interfaces. Applications encode it into data; protocols package and transport it; network devices forward it; the receiving application decodes it for a person or another system. Text, images, audio and video must be represented digitally before transmission.

3.3 TCP/IP Application Layer services and common ports

Protocol	Purpose	Default well-known port(s)	Transport / key point
FTP	File transfer; separate control and data channels.	TCP 21 control; TCP 20 active-mode data (traditional).	TCP; traditional FTP is not encrypted.
HTTP	Web request/response protocol.	TCP 80 (HTTP/1.1 and HTTP/2 typical).	HTTP/3 uses QUIC over UDP 443.
HTTPS	HTTP protected by TLS.	TCP 443 commonly; UDP 443 for HTTP/3 over QUIC.	Confidentiality, integrity, server authentication; certificates.
POP3	Downloads/retrieves email from server.	TCP 110; POP3S 995.	Often download-oriented; client may keep or remove server copy.
IMAP	Accesses and synchronises mailboxes on server.	TCP 143; IMAPS 993.	Folders, flags and synchronisation across devices.
SMTP	Sends/relays email.	TCP 25 server relay; 587 message submission; 465 implicit TLS submission.	Sending, not mailbox retrieval.
DNS	Resolves names and stores DNS records.	UDP 53 commonly; TCP 53 also used.	TCP for zone transfers and some large/truncated responses.
DHCP	Automatically leases IP configuration.	UDP 67 server; UDP 68 client (IPv4).	DORA: Discover, Offer, Request, Acknowledge.

3.4 FTP

FTP is a client-server protocol for file transfer. The control connection typically uses TCP port 21. In active FTP, the server initiates the data connection (traditionally from TCP 20) toward the client; in passive FTP, the client initiates a second connection to a server-advertised port, which works better through many firewalls/NATs. FTP credentials and data are not encrypted by standard FTP. FTPS adds TLS; SFTP is a different file-transfer protocol carried over SSH, commonly TCP 22.

3.5 HTTP and HTTPS

HTTP is a stateless request-response application protocol. A client sends a method, target, headers and optional body; the server returns a status code, headers and optional body. Common methods: GET (retrieve), POST (submit/create processing), PUT (replace/create at known URI), PATCH (partial update), DELETE (remove), HEAD (headers only), OPTIONS (capabilities).

Status class	Meaning	Examples
1xx	Informational	100 Continue

Status class	Meaning	Examples
2xx	Success	200 OK; 201 Created; 204 No Content
3xx	Redirection	301 Moved Permanently; 302 Found; 304 Not Modified
4xx	Client error	400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found
5xx	Server error	500 Internal Server Error; 502 Bad Gateway; 503 Service Unavailable

HTTPS is HTTP protected by TLS. TLS provides encryption in transit, integrity protection and authentication (normally server authentication using a certificate). HTTPS does not guarantee that a website is honest or malware-free; it protects the connection to the authenticated endpoint.

3.6 Email protocols: POP3, IMAP and SMTP

- **SMTP:** sends outgoing mail from client to submission server and relays mail between mail servers.
- **POP3:** retrieves mail, often downloading messages to a client; synchronisation of folders/read state is limited compared with IMAP.
- **IMAP:** keeps mailbox state on the server and synchronises folders, flags and messages across devices.

3.7 DNS

The Domain Name System is a distributed, hierarchical naming system. A resolver asks DNS servers to translate a domain name into records such as IP addresses. Typical lookup chain: client stub resolver → recursive resolver → root server → top-level-domain server → authoritative server. Caching reduces delay and DNS load; cached data is governed by TTL.

Record	Purpose
A	Maps a name to an IPv4 address.
AAAA	Maps a name to an IPv6 address.
CNAME	Alias from one name to another canonical name.
MX	Identifies mail exchanger(s) for a domain; includes priority.
NS	Identifies authoritative name servers for a zone.
PTR	Reverse lookup: address to name.
TXT	Text data; often SPF/DKIM/verification information.
SOA	Start of Authority; zone administrative and timing information.

3.8 DHCP

DHCP automatically provides clients with configuration such as IP address, subnet mask/prefix, default gateway, DNS servers and lease time. IPv4 DORA: Discover (client broadcast), Offer (server proposes), Request (client requests selected offer), Acknowledge (server confirms lease). DHCP relay forwards client broadcasts across routed networks because broadcasts are not normally routed.

Quick revision and MCQ traps

- FTP 21 control; DNS 53; HTTP 80; HTTPS 443; SMTP 25/587; POP3 110; IMAP 143.
- DHCPv4 uses UDP 67 server and UDP 68 client.
- SMTP sends; POP3 and IMAP retrieve/access.
- A maps IPv4; AAAA maps IPv6; MX is mail routing; PTR is reverse lookup.
- HTTPS is HTTP over TLS; a padlock does not certify the site's trustworthiness.

Common traps: DNS can use TCP as well as UDP. • HTTP is stateless by design; cookies/tokens can maintain application state. • SFTP is not simply encrypted FTP; it is a different protocol over SSH.

Q1. Which protocol is primarily used to send email?

A. POP3 **B.** IMAP **C.** SMTP **D.** DNS

Answer: C. SMTP is used for message submission and mail relay.

Q2. Which DNS record maps a hostname to an IPv6 address?

A. A **B.** AAAA **C.** PTR **D.** MX

Answer: B. AAAA stores IPv6 address data.

Q3. What is the correct DHCPv4 sequence?

A. Offer, Discover, ACK, Request **B.** Discover, Offer, Request, ACK **C.** Request, Discover, Offer, ACK **D.** Discover, Request, Offer, ACK

Answer: B. DORA stands for Discover, Offer, Request, Acknowledge.

Q4. Which protocol is designed for server-side mailbox synchronisation?

A. IMAP **B.** FTP **C.** SMTP **D.** TFTP

Answer: A. IMAP maintains mailbox folders and state on the server.

Q5. What does HTTPS add to HTTP?

A. IP routing **B.** TLS security **C.** MAC learning **D.** DHCP leasing

Answer: B. TLS protects HTTP traffic with encryption, integrity and authentication.

4. OSI Transport Layer

4.1 Roles and services

The Transport Layer provides process-to-process communication between applications on end hosts. It segments application data, uses port numbers to identify services, may provide reliability and flow control, and can multiplex many application conversations over one network interface.

- **Segmentation and reassembly:** divides data into transport units and reconstructs it at the destination.
- **Multiplexing/demultiplexing:** allows many applications to share network connectivity using port numbers.
- **Reliability:** TCP uses sequence numbers, acknowledgements, timers and retransmissions.
- **Flow control:** TCP receiver advertises a window to avoid overwhelming its receive buffer.
- **Congestion control:** TCP adjusts sending behaviour to reduce network congestion.

4.2 TCP

Transmission Control Protocol is connection-oriented and reliable. Before normal data transfer, endpoints establish a logical connection using the three-way handshake. TCP provides ordered byte-stream delivery, error detection, retransmission, flow control and congestion control. It does not preserve application message boundaries.

TCP three-way handshake

Client → Server: SYN (initial sequence number)
Server → Client: SYN + ACK
Client → Server: ACK

TCP termination

A typical graceful close uses FIN and ACK exchanges in both directions (often four segments). A RST resets/aborts a connection. TIME_WAIT allows delayed segments to expire and helps ensure the final acknowledgement can be retransmitted if needed.

TCP header fields

Field	Purpose
Source/Destination port	Identify sending and receiving application endpoints.
Sequence number	Numbering of bytes in the stream; supports ordering and retransmission.
Acknowledgment number	Next byte expected from the other side when ACK is set.
Data offset	TCP header length.
Flags	SYN, ACK, FIN, RST, PSH, URG; modern TCP also has ECE/CWR/NS.
Window size	Advertised receive window for flow control.
Checksum	Error detection over TCP header/data and IP pseudo-header.
Options	MSS, window scaling, timestamps and other negotiated features.

4.3 Managing TCP sessions

A TCP connection is identified by a 4-tuple: source IP, source port, destination IP and destination port (protocol is also relevant in broader flow identification). A server commonly listens on a well-known port; clients use ephemeral source ports. TCP is full duplex: both directions can send data independently.

4.4 UDP

User Datagram Protocol is connectionless and message-oriented. It has a small 8-byte header and does not provide built-in delivery guarantees, ordering, retransmission or flow control. Applications choose UDP when low overhead, low latency, multicast support or application-specific reliability is useful.

TCP	UDP
Connection-oriented	Connectionless
Reliable, ordered byte stream	Best-effort datagrams; no guaranteed order
20-byte minimum header	8-byte header
Flow and congestion control	No built-in transport flow/congestion control equivalent
Common: HTTP/1.1, HTTPS over TCP, SSH, SMTP, FTP	Common: DNS queries, DHCP, VoIP, streaming, gaming; QUIC uses UDP

4.5 ICMP

Internet Control Message Protocol supports IP operation by carrying error reports and diagnostic/control messages. It is encapsulated in IP (IPv4 protocol number 1; ICMPv6 uses IPv6 Next Header 58). Ping uses Echo Request and Echo Reply. Traceroute uses TTL/Hop Limit expiry messages (implementation varies). ICMP is not a transport protocol like TCP or UDP and has no TCP/UDP port numbers.

ICMP message	Use
Echo Request / Echo Reply	Reachability/round-trip diagnostic (ping).
Destination Unreachable	Destination/network/host/port or other delivery problem.
Time Exceeded	TTL expired in transit; used by traceroute.
Redirect	Router informs a host of a potentially better next hop (where permitted).

Quick revision and MCQ traps

- TCP is reliable, connection-oriented and ordered; UDP is connectionless and lightweight.
- TCP handshake is SYN → SYN-ACK → ACK.
- TCP header minimum is 20 bytes; UDP header is 8 bytes.
- Ports identify processes; ICMP uses message types/codes, not TCP/UDP ports.

- Ping tests IP reachability, not whether a particular application service is working.

Common traps: UDP checksum behaviour differs between IPv4 and IPv6; do not claim UDP never checks errors. • A successful ping does not prove TCP port 443 is open. • TCP is a byte stream, not a message-preserving protocol.

Q1. Which TCP flag starts a connection?

A. FIN **B.** RST **C.** SYN **D.** PSH

Answer: C. SYN synchronises initial sequence numbers during connection establishment.

Q2. What is the minimum TCP header size without options?

A. 8 bytes **B.** 20 bytes **C.** 32 bytes **D.** 60 bytes

Answer: B. TCP minimum header is 20 bytes.

Q3. Which protocol has an 8-byte fixed header?

A. TCP **B.** UDP **C.** ICMP **D.** HTTP

Answer: B. UDP has an 8-byte header.

Q4. Which ICMP message is associated with traceroute?

A. Echo Reply **B.** Time Exceeded **C.** Redirect only **D.** Source Quench

Answer: B. A router returns Time Exceeded when TTL/Hop Limit expires.

Q5. A TCP connection is commonly identified by which values?

A. Only destination MAC **B.** Source/destination IP and source/destination ports **C.** Only DNS name **D.** VLAN ID and cable type

Answer: B. The four-tuple identifies the endpoint pair and ports.

5. OSI Network Layer

5.1 Responsibilities

The Network Layer (Layer 3) provides logical addressing and moves packets between different networks. IP is connectionless and best-effort: it does not guarantee delivery, order or recovery. Routers examine the destination IP address and select a route using their routing table.

5.2 IPv4 address structure

An IPv4 address is 32 bits, written as four decimal octets separated by dots (for example, 192.168.10.25). Each octet ranges from 0 to 255. A subnet mask/prefix identifies which bits represent the network prefix and which identify hosts.

Term	Meaning
Network address	Address with all host bits set to 0; identifies the subnet.
Broadcast address	Address with all host bits set to 1; reaches all hosts on that IPv4 subnet.
Host address	Usable unicast address assigned to an interface, excluding network and broadcast in ordinary subnets.
Prefix length	Number of leading network bits, e.g. /24.
Default gateway	Router interface used to reach destinations outside the local subnet.

5.3 IPv4 address classes (historical classful system)

Class	First octet	Default mask	Historical use
A	1–126	255.0.0.0 (/8)	Large networks; 0 and 127 are special.
B	128–191	255.255.0.0 (/16)	Medium networks.
C	192–223	255.255.255.0 (/24)	Small networks.
D	224–239	Not for ordinary unicast subnetting	Multicast.
E	240–255	Not ordinary unicast	Experimental/reserved.

Modern routing uses CIDR/classless prefixes rather than default class boundaries. 127.0.0.0/8 is loopback. Private IPv4 ranges: 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. 169.254.0.0/16 is IPv4 link-local/APIPA.

5.4 IPv6 overview

IPv6 addresses are 128 bits and written as eight groups of hexadecimal hextets separated by colons. Leading zeros in a hextet may be omitted; one continuous run of zero hextets may be compressed once using ::. IPv6 has no broadcast; it uses multicast and anycast. Common types include global unicast (commonly 2000::/3), link-local (FE80::/10), unique local (FC00::/7) and loopback (::1).

5.5 Grouping devices and hierarchical addressing

Subnetting divides a larger address block into smaller logical networks. Hierarchical addressing makes routing scalable: routers can summarise contiguous prefixes, reducing routing-table size. A host decides whether a destination is local by applying its subnet mask to its own IP and the destination IP; if local, it sends directly at Layer 2, otherwise it sends to the default gateway.

5.6 Routes, next hop and packet forwarding

- **Connected route:** automatically installed for an active interface with an IP address.
- **Static route:** manually configured by an administrator.
- **Dynamic route:** learned through a routing protocol such as RIP, OSPF or EIGRP.
- **Next hop:** IP address of the next router toward the destination, or an exit interface.
- **Longest Prefix Match:** router chooses the matching route with the most specific (longest) prefix.
- **Default route:** route used when no more specific route matches; IPv4 0.0.0.0/0.

At each hop, a router decrements IPv4 TTL (or IPv6 Hop Limit), checks the destination against its routing table, resolves the next-hop Layer 2 address if needed, and forwards a newly encapsulated frame. If no route exists and no default route applies, the packet is dropped; an ICMP error may be sent.

5.7 Subnetting: formulas

Quantity	Formula / rule	Notes
Number of subnets created	2^s	s = bits borrowed from host portion.
Addresses per subnet	2^h	h = remaining host bits.
Usable IPv4 hosts (ordinary subnet)	2^h – 2	Subtract network and broadcast; exceptions exist for /31 point-to-point and /32 host routes.
Prefix length	/ (32 – h)	IPv4 has 32 total bits.
Block size in interesting octet	256 – mask value	For example, /26 mask 255.255.255.192 gives block size 64.

5.8 Solved subnetting examples

Example 1: 192.168.1.0/24 divided into /26

Borrow 2 bits from the /24 host portion, giving four /26 subnets. Each /26 has 64 total addresses and normally 62 usable hosts. Mask = 255.255.255.192; block size = 256 – 192 = 64.

Subnet	Network	Usable host range	Broadcast
1	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Example 2: Find subnet details for 10.10.5.77/27

A /27 mask is 255.255.255.224. Block size = 256 – 224 = 32. The last-octet ranges are 0–31, 32–63, 64–95, 96–127, etc. Since 77 falls in 64–95: network = 10.10.5.64; broadcast = 10.10.5.95; usable range = 10.10.5.65–10.10.5.94; usable hosts = 30.

FLSM versus VLSM

FLSM	VLSM
All subnets use the same prefix/mask.	Different subnets can use different prefix lengths.
Simple planning and administration.	More efficient address utilisation.
Can waste addresses when subnet needs differ.	Requires careful allocation and non-overlapping blocks.
Use when subnet sizes are similar.	Use when departments/links need different host counts.

VLSM method: list subnet requirements from largest to smallest; choose the smallest prefix that supports each requirement; allocate aligned blocks; ensure no overlap; reserve growth where possible. For n required usable hosts, select h such that 2^h – 2 ≥ n (ordinary subnet), then prefix = 32 – h.

Quick revision and MCQ traps

- IPv4 is 32 bits; IPv6 is 128 bits.
- Network = host bits all zero; broadcast = host bits all one.
- Private IPv4 blocks are 10/8, 172.16/12 and 192.168/16.
- Router chooses longest prefix match.
- Default route is 0.0.0.0/0 for IPv4.
- FLSM uses equal-size subnets; VLSM uses variable-size subnets.

Common traps: Do not subtract 2 for IPv6; IPv6 has no broadcast address. • /24 has 256 total addresses and normally 254 usable hosts. • A /30 has 4 total and 2 usable IPv4 addresses; /31 is a point-to-point exception. • Classful A/B/C defaults are historical; CIDR is modern.

Q1. How many usable host addresses are normally available in a /27 IPv4 subnet?

A. 14 **B.** 30 **C.** 32 **D.** 62

Answer: B. A /27 leaves 5 host bits: 2^5 – 2 = 30 usable hosts.

Q2. Which is a private IPv4 range?

A. 11.0.0.0/8 **B.** 172.16.0.0/12 **C.** 192.0.2.0/24 **D.** 224.0.0.0/4

Answer: B. 172.16.0.0 through 172.31.255.255 is private.

Q3. What does a router use to select the most specific route?

A. Lowest MAC address **B.** Longest prefix match **C.** Highest TTL **D.** DNS priority

Answer: B. The route with the longest matching network prefix wins.

Q4. What is the IPv4 broadcast address for 192.168.1.64/26?

A. 192.168.1.64 **B.** 192.168.1.126 **C.** 192.168.1.127 **D.** 192.168.1.255

Answer: C. The /26 block spans 64–127; 127 is the broadcast address.

Q5. Which statement about IPv6 is correct?

A. It uses 32-bit addresses **B.** It supports broadcast **C.** It uses 128-bit addresses **D.** It is written only in decimal

Answer: C. IPv6 addresses are 128 bits and written in hexadecimal.

6. OSI Data Link Layer

6.1 Responsibilities and services

The Data Link Layer (Layer 2) transfers frames across one local link. It organises bits into frames, uses physical/MAC addresses, controls access to shared media and detects certain transmission errors. It does not normally provide end-to-end delivery across multiple routed networks.

- Framing: marks the beginning and end of a data unit.
- Physical addressing: source and destination MAC addresses in Ethernet/Wi-Fi frames.
- Media access control: decides how devices share a medium.
- Error detection: Frame Check Sequence (FCS) using CRC detects corruption; it does not itself correct every error.
- Link-level flow control may be present in some technologies.

6.2 Sublayers

Sublayer	Role
LLC (Logical Link Control)	Interfaces with Network Layer and identifies/multiplexes network-layer protocols; historically associated with IEEE 802.2.
MAC (Media Access Control)	Controls access to the medium and handles MAC addressing and frame format.

6.3 MAC addresses

A common Ethernet MAC address is 48 bits (6 bytes), written in hexadecimal, for example 00:1A:2B:3C:4D:5E. The first 24 bits commonly identify the vendor-assigned OUI (Organizationally Unique Identifier); the remaining bits identify the interface within that assignment. The least significant bit of the first octet indicates individual/group (I/G): 0 = individual/unicast, 1 = group/multicast. The next bit indicates universal/local administration.

FF:FF:FF:FF:FF:FF is the Ethernet broadcast destination. A switch learns the source MAC address and ingress port, then forwards based on destination MAC. Unknown unicast and broadcast frames are flooded within the VLAN (excluding ingress port); known unicast is sent to the learned port.

6.4 Ethernet and other Data Link protocols

Technology/protocol	Purpose / key facts
Ethernet (IEEE 802.3)	Dominant wired LAN technology; frame includes destination/source MAC, type/length, payload and FCS.
Wi-Fi MAC (IEEE 802.11)	Wireless LAN link layer; uses association and radio medium access (CSMA/CA).
PPP	Point-to-point data-link protocol; supports link encapsulation and optional authentication.
HDLC	Bit-oriented data-link protocol used on point-to-point/serial links; Cisco HDLC is vendor-specific.
ARP (IPv4)	Resolves a local IPv4 next-hop address to a MAC address; operates between Layer 2 and Layer 3.
VLAN tagging (802.1Q)	Adds VLAN identification to Ethernet frames on trunk links.

6.5 Copper and fibre; Ethernet cable types

Twisted-pair copper carries electrical signals and is common in Ethernet access cabling. Fibre-optic cable carries light, supports long distances and is immune to electromagnetic interference. Multimode fibre is generally used for shorter distances; single-mode fibre supports longer distances. Actual reach depends on transceiver, speed and cable specification.

Cable / medium	Typical use	Important point
UTP copper	Common Ethernet LAN links.	Twisted pairs reduce crosstalk; connector commonly 8P8C (often called RJ-45).
Straight-through Ethernet	Traditionally connects unlike devices (PC-to-switch, router-to-switch).	Modern auto-MDI/MDIX often removes the need to choose cable type.
Crossover Ethernet	Traditionally connects like devices (switch-to-switch, PC-to-PC, router-to-router Ethernet).	Swaps transmit/receive pairs for 10/100 Ethernet; Gigabit uses all pairs and modern devices auto-negotiate.
Fibre optic	Backbone, uplinks, long distance/high bandwidth.	Uses optical transceivers/connectors; not affected by EMI.

Quick revision and MCQ traps

- Layer 2 PDU is a frame; Layer 3 PDU is a packet.
- MAC address is commonly 48 bits; IPv4 address is 32 bits.
- Switch learns from source MAC, forwards using destination MAC.
- LLC and MAC are Data Link sublayers.
- Ethernet FCS uses CRC for error detection.

Common traps: A MAC address is not a substitute for an IP address across routed networks. • A switch floods unknown unicast only within the relevant VLAN, not to every network. • Straight-through/crossover rules are traditional; auto-MDI/MDIX changes practical cabling requirements.

Q1. Which Data Link sublayer controls access to the transmission medium?

A. LLC **B.** MAC **C.** IP **D.** TCP

Answer: B. MAC handles medium access and MAC addressing.

Q2. How long is a standard Ethernet MAC address?

A. 32 bits **B.** 48 bits **C.** 64 bits **D.** 128 bits

Answer: B. A conventional Ethernet MAC address is 48 bits.

Q3. Which address does a switch learn from an incoming frame?

A. Destination IP **B.** Source MAC **C.** TCP port **D.** DNS name

Answer: B. Switches populate their MAC table using the source MAC and ingress port.

Q4. What is the Ethernet broadcast MAC address?

A. 00:00:00:00:00:00 **B.** FF:FF:FF:FF:FF:FF **C.** 01:00:5E:00:00:00 **D.** AA:AA:AA:AA:AA:AA

Answer: B. All bits set to 1 in the 48-bit destination MAC.

Q5. What is the primary role of Ethernet FCS?

A. Encryption **B.** Error detection **C.** Routing **D.** Address allocation

Answer: B. FCS/CRC detects frame corruption.

7. OSI Physical Layer

7.1 Functions and services

The Physical Layer (Layer 1) transmits raw bits over a medium. It defines electrical, optical or radio characteristics; connectors and pinouts; signal encoding/modulation; bit timing; data rates; transmission modes; and physical topology. It does not interpret MAC or IP addresses.

7.2 Physical signalling sublayer

Digital data must be represented as signals. Copper uses electrical voltage transitions; fibre uses light; wireless uses radio waves. Encoding maps bits to signal patterns; modulation changes a carrier's amplitude, frequency or phase to carry information. Clocking/synchronisation allows receiver and sender to identify bit boundaries.

7.3 Transmission concepts

Concept	Meaning
Bandwidth	The capacity/range of a channel; in digital networking often expressed as bit/s, though technically bandwidth can refer to frequency range.
Throughput	Actual achieved data rate.
Goodput	Useful application payload rate excluding protocol overhead and retransmissions.
Latency	Time taken for data to travel; often measured round-trip or one-way.
Jitter	Variation in packet delay; important for real-time voice/video.
Attenuation	Signal strength reduction over distance.
Noise/interference	Unwanted energy that distorts a signal.
Duplex	Half duplex sends both directions but not simultaneously; full duplex sends simultaneously.

7.4 Physical media

Medium	Benefits	Limitations / examples
UTP copper	Low cost, easy installation, common office Ethernet.	Distance and EMI limitations; twisted pairs mitigate interference.
STP copper	Additional shielding can reduce EMI.	Costlier and grounding/installation matter.
Multimode fibre	Good for shorter building/campus links; larger core.	Distance lower than single-mode at comparable systems.
Single-mode fibre	Long-distance and high-capacity backbone links.	Optics and installation can cost more.
Wireless radio	Mobility and easy deployment.	Shared spectrum, interference, security and coverage challenges.

Quick revision and MCQ traps

- Layer 1 carries bits; it defines media, connectors, signalling and timing.
- Throughput is actual achieved rate; bandwidth is capacity; goodput is useful payload rate.
- Jitter is variation in delay; latency is delay itself.
- Fibre uses light and is immune to electromagnetic interference.

Common traps: A high bandwidth link can still have high latency. • Full duplex is simultaneous two-way communication; half duplex is not. • Physical Layer does not perform IP routing or MAC learning.

Q1. Which layer defines connectors, signal levels and bit transmission?

A. Application **B.** Network **C.** Data Link **D.** Physical

Answer: D. These are Physical Layer responsibilities.

Q2. Variation in packet delay is called:

A. Attenuation **B.** Jitter **C.** Throughput **D.** Encapsulation

Answer: B. Jitter is variation in delay.

Q3. Which medium is immune to electromagnetic interference?

A. UTP **B.** Coaxial copper **C.** Fibre optic **D.** Unshielded serial copper

Answer: C. Fibre transmits light rather than electrical signals.

Q4. What is goodput?

A. Nominal link capacity **B.** Useful application data delivered per unit time **C.** Maximum cable length **D.** Round-trip time

Answer: B. Goodput excludes protocol overhead and retransmitted data.

8. TCP/IP Protocol Suite

The TCP/IP suite is the family of protocols used for communication across the Internet and most enterprise networks. It is not one protocol: it includes application protocols (HTTP, DNS, SMTP), transport protocols (TCP, UDP), Internet protocols (IPv4, IPv6, ICMP) and link technologies (Ethernet, Wi-Fi, PPP).

Layer	Representative protocols/technologies	What it provides
Application	HTTP(S), DNS, DHCP, FTP, SMTP, POP3, IMAP, SSH, SNMP	Services for user applications and network management.
Transport	TCP, UDP; QUIC over UDP	Process-to-process delivery and transport behaviour.
Internet	IPv4, IPv6, ICMP, ICMPv6	Logical addressing and packet routing/control.
Network Access	Ethernet, Wi-Fi, PPP, ARP (IPv4 support)	Local framing, media access and physical transmission.

How a web request travels

1. User enters a URL; browser determines the hostname and needs an IP address.
2. DNS resolver obtains an address (from cache or DNS hierarchy).
3. Host checks whether destination is local; otherwise it sends to its default gateway.
4. Transport connection is established (TCP handshake for traditional HTTPS over TCP; QUIC setup for HTTP/3).
5. TLS authenticates the server and establishes cryptographic protection for HTTPS.
6. HTTP request is sent; routers forward packets hop by hop.
7. Server processes the request and returns an HTTP response; browser renders content.

EXAM ALERT: TCP/IP is a protocol suite. TCP is one transport protocol within that suite; IP is a Network/Internet Layer protocol.

9. Security: Firewalls and Proxies

9.1 Security foundations

Network security commonly aims for the CIA triad: **Confidentiality** (only authorised parties can read data), **Integrity** (data is not improperly altered) and **Availability** (systems/services remain accessible). Authentication verifies identity; authorisation determines permitted actions; accounting/auditing records activity.

9.2 Firewall: purpose and operation

A firewall enforces a security policy by allowing, denying or inspecting traffic between interfaces, hosts or zones. Rules may use source/destination IP, protocol, ports, connection state, application identity or user identity. Firewalls can be network-based or host-based.

Firewall type	How it works	Use / limitation
Packet-filtering (stateless)	Checks packet headers against rules.	Fast and simple; limited awareness of connection context.
Stateful inspection	Tracks connection state and permits packets belonging to valid sessions.	Common enterprise firewall behaviour.
Circuit-level gateway	Validates sessions/connection establishment rather than inspecting full application content.	Can hide internal session details; limited content inspection.
Application proxy / gateway	Terminates and proxies application traffic; can inspect Layer 7 content.	Detailed control; may add latency and require application support.

Firewall type	How it works	Use / limitation
Next-generation firewall (NGFW)	Combines stateful firewall with application awareness and often IDS/IPS, URL or malware controls.	Capabilities vary by vendor/licence.
Host-based firewall	Runs on an endpoint/server and filters that host's traffic.	Protects even when device leaves the corporate network.

9.3 Firewall deployment examples

- Allow users to access public HTTPS websites while blocking unsolicited inbound connections.
- Permit only approved management sources to reach SSH on network devices.
- Separate guest Wi-Fi from internal business systems.
- Publish a web server in a DMZ while restricting its access to internal databases.
- Log denied traffic and alert on suspicious patterns.

9.4 Proxy server: purpose and types

A proxy is an intermediary that makes requests on behalf of a client or server. A forward proxy represents clients; a reverse proxy represents servers.

Proxy type	Position and purpose	Example
Forward proxy	Between internal clients and external services; applies browsing policy, filtering, caching or identity controls.	Company web proxy.
Reverse proxy	In front of backend servers; routes requests, terminates TLS, hides backend details and may load-balance.	Public website reverse proxy.
Transparent proxy	Intercepts traffic with little/no explicit client configuration (implementation dependent).	Managed network content filtering.
Caching proxy	Stores reusable responses to reduce bandwidth and response time.	Repeated software/document downloads.
Web application proxy / gateway	Understands HTTP(S) and can enforce application-level rules.	WAF-like request filtering.

9.5 Firewall versus proxy

Firewall	Proxy
Primarily enforces traffic policy between networks/hosts.	Acts as an intermediary endpoint for client/server requests.
May inspect L3/L4 and, in advanced products, L7.	Often operates at application layer for web traffic.
Can allow/deny without terminating application sessions.	Typically terminates one connection and creates another.
Protects boundaries and endpoints.	Can provide filtering, caching, identity mediation and server shielding.

9.6 NAC overview

Network Access Control (NAC) determines whether a device/user may connect and what level of access it receives. NAC may check identity, device posture, patch status, endpoint protection and compliance, then place the endpoint into an appropriate VLAN or restricted network. It can use 802.1X, RADIUS and policy systems. NAC is not the same as NAT: NAC controls admission; NAT translates addresses.

Quick revision and MCQ traps

- Firewall enforces policy; proxy intermediates requests.
- Forward proxy represents clients; reverse proxy represents servers.
- Stateful firewalls track sessions; stateless filters inspect individual packets.

- DMZ isolates publicly reachable services from trusted internal networks.
- NAC evaluates who/what connects and assigns access policy.

Common traps: A proxy is not automatically a firewall, although products can combine functions. • HTTPS encryption can limit inspection unless the security architecture legitimately terminates/decrypts TLS. • NAT is address translation, not an access-control system by itself.

Q1. Which firewall tracks active connection state?

A. Stateless packet filter **B.** Stateful inspection firewall **C.** Repeater **D.** Hub

Answer: B. Stateful inspection maintains session information.

Q2. A proxy placed in front of web servers is called:

A. Forward proxy **B.** Reverse proxy **C.** Transparent bridge **D.** DHCP relay

Answer: B. A reverse proxy represents backend servers to clients.

Q3. Which security goal ensures data is not altered improperly?

A. Availability **B.** Integrity **C.** Confidentiality **D.** Scalability

Answer: B. Integrity protects against unauthorised modification.

Q4. What is the main purpose of NAC?

A. Translate private addresses **B.** Control network admission based on identity/posture **C.** Resolve domain names **D.** Encrypt all Ethernet frames

Answer: B. NAC evaluates endpoints/users and applies access policy.

10. Web Services and HTTP/HTTPS

A web service exposes functionality or data over a network through a defined interface. A client sends a request to an endpoint; the service authenticates/authorises it, processes the request and returns a response, commonly in JSON or XML. Web services enable different platforms and programming languages to interoperate.

10.1 Common web-service styles

Style	Characteristics
REST	Architectural style using resources, URIs and HTTP methods; commonly JSON; stateless interactions are typical.
SOAP	Protocol with XML message envelope and formal service contracts; can use HTTP and other transports.
RPC-style APIs	Client invokes remote procedures/functions; may use JSON-RPC, XML-RPC or gRPC.
WebSocket	Persistent, full-duplex communication channel over a connection, useful for real-time updates.

10.2 Typical request-response flow

1. Client discovers or knows service endpoint URL.
2. Client resolves the server name using DNS.
3. Client establishes transport connectivity and, for HTTPS, TLS protection.
4. Client sends HTTP method, path, headers and body (if applicable).
5. Server validates authentication, authorisation, input and business rules.
6. Server returns status code, headers and response body.

EXAM ALERT: HTTP is an application-layer protocol. HTTPS is HTTP protected by TLS; TLS is not itself an HTTP method or status code.

Q1. Which format is commonly used by REST APIs?

A. JSON **B.** BIOS **C.** FAT32 **D.** ARP

Answer: A. JSON is widely used, although REST is not limited to JSON.

Q2. Which HTTP method is generally used to retrieve a resource?

A. DELETE **B.** GET **C.** PATCH **D.** POST

Answer: B. GET requests representation of a resource.

Q3. What does HTTP 404 indicate?

A. Success **B.** Not Found **C.** Server timeout only **D.** Permanent redirect

Answer: B. 404 is a client-error status indicating the requested resource was not found.

PART B — CISCO PACKET TRACER AND NETWORK ADMINISTRATION

11. Recap of Network Basics and Addressing

11.1 Packet Tracer overview

Cisco Packet Tracer is a network simulation and learning tool. It lets learners place virtual routers, switches, PCs, servers and links; configure devices using a CLI or GUI; inspect packet flow in Simulation mode; and practise topology design. It models many networking behaviours but is not a perfect replacement for physical equipment or every real IOS feature.

11.2 Packet Tracer interface and modes

- **Realtime mode:** network runs continuously and events happen as configured.
- **Simulation mode:** packet events can be inspected step by step; filters help focus on protocols.
- **Logical workspace:** build and arrange the topology.
- **Physical workspace:** view physical locations/racks and device placement where supported.

11.3 Address types and unicast/broadcast/multicast

A network address identifies a subnet, a broadcast address targets all IPv4 hosts on that subnet, and a host address is assigned to an interface. Unicast is one-to-one, broadcast is one-to-all in a broadcast domain, and multicast is one-to-many members of a group.

11.4 Subnetting practice

Use the subnetting method in Chapter 5. In Packet Tracer, configure IP address and mask on each PC, configure router interface addresses, enable interfaces with `no shutdown`, and verify with `ipconfig`, `ping` and router `show ip interface brief`.

11.5 FLSM and VLSM planning example

Suppose a site needs networks for 50, 25, 12 and 2 hosts. With VLSM, allocate largest first: 50 hosts needs /26 (62 usable); 25 needs /27 (30 usable); 12 needs /28 (14 usable); a point-to-point link can use /30 (2 usable) or /31 where supported. Allocate each block on correct boundaries and document gateway, network and broadcast addresses.

EXAM ALERT: For a subnet needing 50 usable hosts, /26 works because 2^6 – 2 = 62; /27 fails because 2^5 – 2 = 30.

12. Cisco CLI and Basic Device Administration

12.1 CLI modes

Mode	Prompt example	Purpose / command to enter
User EXEC	Router>	Basic monitoring; enter enable.
Privileged EXEC	Router#	Verification and management; enter configure terminal.
Global configuration	Router(config)#	Device-wide configuration.
Interface configuration	Router(config-if)#	Configure a specific interface.
Line configuration	Router(config-line)#	Configure console, AUX or VTY lines.
Router configuration	Router(config-router)#	Configure routing protocol process.

12.2 Essential navigation and help

Command	Purpose
?	Context-sensitive help for available commands.
show ?	List available show commands.
Tab	Complete a partially typed command.
Ctrl+A / Ctrl+E	Move to beginning/end of command line (terminal dependent).
Ctrl+C	Abort current command or escape configuration prompt in many contexts.
exit	Move back one configuration level / exit a mode.
end or Ctrl+Z	Return to privileged EXEC from configuration mode.
do show ...	Run an EXEC show command from configuration mode.

12.3 Hostname, passwords and banners

```
enable
configure terminal
hostname R1
enable secret StrongSecret
line console 0
password ConsolePass
login
exit
line vty 0 4
password VtyPass
login
exit
banner motd #Authorised access only#
end
```

enable secret protects privileged EXEC access and is stored using a stronger password-hashing mechanism than legacy **enable password**. If both are configured, enable secret takes precedence. Console and VTY line passwords require `login` to prompt for the configured line password. For stronger modern access, prefer local usernames and SSH rather than Telnet or shared line passwords.

12.4 Password encryption and configuration saving

```
service password-encryption
copy running-config startup-config
show running-config
show startup-config
```

`service password-encryption` obfuscates certain plaintext line passwords in configuration; it is reversible weak encryption and is not a substitute for `enable secret` or strong authentication. Running-config is active in RAM and is lost on reload if not saved. Startup-config is the saved configuration in NVRAM (platform dependent).

12.5 IOS and configuration terms

Term	Meaning
IOS / IOS image	Cisco network operating system software image.
RAM	Holds running configuration, routing tables and active operational data; volatile.
NVRAM	Commonly stores startup configuration; non-volatile.
Flash	Stores IOS image and files; non-volatile.
ROM	Bootstrap/diagnostic code on many platforms.
Running-config	Current active configuration.
Startup-config	Configuration loaded during boot.

Quick revision and MCQ traps

- User EXEC prompt ends with >; privileged EXEC ends with #.
- enable enters privileged EXEC; configure terminal enters global configuration.
- hostname changes device prompt/name.
- enable secret is preferred over enable password.
- copy running-config startup-config saves changes.

Common traps: A command entered in the wrong CLI mode may be rejected. • service password-encryption is not strong cryptographic protection. • Saving is essential: running-config and startup-config are not the same.

Q1. Which prompt indicates privileged EXEC mode?

A. R1> **B.** R1# **C.** R1(config)# **D.** R1(config-if)#

Answer: B. The # prompt indicates privileged EXEC mode.

Q2. Which command saves active configuration for the next boot?

A. erase startup-config **B.** copy startup-config running-config **C.** copy running-config startup-config **D.** reload

Answer: C. It copies the active running configuration to startup-config.

Q3. Which command is preferred for the privileged EXEC password?

A. enable secret **B.** banner motd **C.** hostname **D.** show version

Answer: A. enable secret stores a protected secret and takes precedence over enable password.

Q4. Which key commonly displays context-sensitive help?

A. ? **B.** Esc **C.** F1 only **D.** Ctrl+S

Answer: A. Question mark displays context-sensitive CLI help.

13. OSI Model Labs, Cables and Connectivity

13.1 Physical layer lab

Select an appropriate cable and interface, connect devices, check link lights and verify interface status. Traditional Packet Tracer exercises use straight-through cable for PC-to-switch and router-to-switch, and crossover for like Ethernet devices. Modern auto-MDI/MDIX may allow either cable on real equipment.

13.2 Layer 2 MAC table commands

Command	Purpose
show mac address-table	Display learned/static MAC address table.
show mac address-table dynamic	Show dynamically learned entries.
show interfaces status	Show switch port status, VLAN and speed/duplex (model dependent).
clear mac address-table dynamic	Clear dynamically learned entries (syntax may vary).
show interfaces	Detailed interface statistics and status.

To observe learning: clear dynamic entries, generate traffic (for example, ping between hosts), then inspect the MAC table. A switch learns the source MAC on ingress. The first ping may time out while ARP resolves addresses.

13.3 Layer 3 interface and loopback configuration

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface loopback 0
ip address 10.10.10.1 255.255.255.255
end
show ip interface brief
```

A loopback is a logical software interface, not a physical port. It normally remains up while the device is operational and is often used for router IDs, management and testing. A physical router interface is administratively down until enabled with `no shutdown`.

13.4 TCP and UDP headers

Packet Tracer can display protocol layers in Simulation/PDU details. Compare source/destination ports, TCP flags and sequence/acknowledgment values with UDP's simpler header. TCP handshake packets show SYN, SYN-ACK and ACK.

13.5 Application services lab

In a Packet Tracer server, open the Services tab and enable the required service (HTTP, DNS, DHCP, FTP, email). Configure server IP/default gateway first. For DNS, add name-to-address records; for DHCP, create a pool with network, mask, default gateway and DNS server; for email, configure domain/user accounts and clients. Ensure routing and ACLs allow the relevant traffic.

EXAM ALERT: Application services fail if lower layers are not working. Verify physical link → VLAN/MAC → IP/subnet/gateway → route → service/port → client settings.

14. Remote Access: Telnet and SSH

14.1 Telnet

Telnet provides remote terminal access, commonly TCP port 23. It transmits session data, including credentials, without encryption. It should be avoided on untrusted networks and is mainly useful in isolated labs or legacy environments.

14.2 SSH

Secure Shell provides encrypted remote login and command execution, commonly TCP port 22. SSH protects credentials and session data and supports host authentication. Cisco IOS SSH setup commonly requires a hostname, domain name, RSA key pair, local username and VTY lines configured for SSH.

```
configure terminal
hostname R1
ip domain-name lab.local
```

```
username admin secret StrongPass123
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
login local
transport input ssh
end
show ip ssh
```

Client connection example: `ssh -l admin 192.168.1.1`. Key size support depends on IOS/platform; Packet Tracer may accept smaller modulus values than current production recommendations.

Telnet	SSH
TCP 23	TCP 22
Plaintext session	Encrypted session
Weak for modern administration	Preferred for secure remote administration
No strong server identity protection by default	Uses host keys and can authenticate users.

Quick revision and MCQ traps

- SSH uses TCP port 22; Telnet uses TCP port 23.
- Use login local with a local username database.
- Use transport input ssh to disable Telnet on VTY lines.

Common traps: Setting a VTY password alone does not configure SSH. • SSH needs crypto keys and appropriate identity configuration on common IOS platforms.

Q1. Which protocol encrypts remote terminal sessions?

A. Telnet **B.** SSH **C.** TFTP **D.** HTTP

Answer: B. SSH encrypts remote terminal traffic.

Q2. What is the default Telnet TCP port?

A. 21 **B.** 22 **C.** 23 **D.** 25

Answer: C. Telnet uses TCP port 23.

Q3. Which command tells VTY lines to authenticate against local usernames?

A. login local **B.** enable secret **C.** service dhcp **D.** ip route

Answer: A. login local uses the local username database.

15. Routing: Static and Dynamic

15.1 Routing fundamentals

Routing is the Layer 3 process of selecting a path and forwarding packets between networks. A router maintains a routing table with destination prefixes, next hops, exit interfaces, metrics and route sources. Connected routes appear when interfaces are operational and addressed.

15.2 Static routing

A static route is manually configured and does not adapt automatically to topology changes. It is predictable, has low protocol overhead and is useful for stub networks, default routes and small stable designs. Its disadvantages are manual maintenance and limited automatic failover unless multiple routes are configured.

```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
ip route 0.0.0.0 0.0.0.0 10.0.0.2
show ip route
show ip route static
```

The first command routes 192.168.2.0/24 via next-hop 10.0.0.2. The second is an IPv4 default route. A floating static route uses a higher administrative distance than the preferred route.

15.3 Dynamic routing

Dynamic routing protocols exchange reachability information and recalculate paths when topology changes. They differ in algorithm, metric, convergence, scalability and resource use.

Protocol	Type / algorithm	Metric / key facts	Typical use
RIP	Distance-vector; hop-count metric.	Maximum 15 hops; 16 means unreachable; periodic updates (RIPv2 supports classless routing and authentication).	Small/simple networks and teaching labs.
EIGRP	Advanced distance-vector / hybrid characteristics; DUAL algorithm.	Composite metric commonly based on bandwidth and delay; feasible successor can provide loop-free backup.	Cisco-originated, now documented for broader implementation; enterprise networks.
OSPF	Link-state; SPF (Dijkstra) algorithm.	Cost metric commonly based on reference bandwidth; areas reduce LSDB scope; Area 0 is backbone.	Medium/large enterprise networks; fast convergence and scalable hierarchy.

15.4 Administrative Distance (AD)

Administrative Distance ranks the trustworthiness of route sources on a Cisco router; lower is preferred when the same prefix is learned from different sources. Common Cisco defaults: connected 0, static 1, EIGRP summary 5, eBGP 20, EIGRP internal 90, OSPF 110, IS-IS 115, RIP 120, EIGRP external 170, iBGP 200. Values are vendor-specific conventions, not universal protocol properties.

15.5 RIP configuration

```
router rip
version 2
no auto-summary
network 10.0.0.0
network 192.168.1.0
end
show ip protocols
show ip route rip
```

RIP's network command enables RIP on matching interfaces and advertises connected networks according to RIP rules. RIPv2 supports subnet masks, multicast updates to 224.0.0.9 and authentication options. RIP uses hop count; 15 is maximum reachable hop count.

15.6 EIGRP configuration (classic IPv4 example)

```
router eigrp 100
network 10.0.0.0 0.0.0.255
network 192.168.1.0 0.0.0.255
no auto-summary
end
show ip eigrp neighbors
show ip route eigrp
```

100 is the autonomous system number and must match among classic EIGRP neighbours. Wildcard masks identify matching interface ranges. Named EIGRP configuration exists on newer IOS.

15.7 OSPF single-area configuration

```
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 192.168.1.0 0.0.0.255 area 0
end
show ip ospf neighbor
show ip route ospf
show ip ospf interface brief
```

OSPF process ID is locally significant and need not match between routers. Neighbours must have compatible area, timers, authentication and network settings. Router ID is a 32-bit value, often chosen manually or from loopback/highest interface address.

15.8 Route redistribution

Redistribution injects routes learned by one routing source into another routing protocol. It is needed when parts of a network use different protocols. It can create routing loops, suboptimal paths and metric problems. Use explicit route filters, tags, metric planning and a clear boundary design. OSPF requires appropriate external route types/metrics; EIGRP needs a seed metric when redistributing routes into EIGRP.

```
router eigrp 100
redistribute ospf 1 metric 10000 100 255 1 1500

router ospf 1
redistribute eigrp 100 subnets
```

The example is illustrative; the EIGRP metric components are bandwidth (Kbit/s), delay (tens of microseconds), reliability, load and MTU. Exact syntax and defaults vary. In a real network, filter what is redistributed and prevent feedback.

15.9 Routing verification and troubleshooting

- `show ip route`: verify route exists and identify source/next hop.
- `show ip interface brief`: check interface status and IP address.
- `show ip protocols`: inspect routing protocol configuration/timers/networks.
- `show ip ospf neighbor` / `show ip eigrp neighbors`: check adjacencies.
- `ping` tests reachability; `traceroute` reveals hop path.
- Check subnet masks, passive interfaces, ACLs, mismatched areas/AS numbers, timers and missing return routes.

Quick revision and MCQ traps

- Static routes are manually configured; dynamic routes are learned.
- RIP metric is hop count; 15 reachable, 16 infinity.
- OSPF is link-state and uses areas; Area 0 is backbone.
- EIGRP uses DUAL; classic EIGRP neighbours share an AS number.
- Lower Cisco AD is preferred; OSPF 110, RIP 120, EIGRP internal 90.
- Redistribution needs careful metric and loop control.

Common traps: Administrative Distance is not the same as routing metric. • OSPF process ID need not match between routers; area must match on a link. • RIP's 16 is infinity, not 16 usable hops. • A route can exist in the table but traffic can still fail due to missing return path or ACL.

Q1. Which routing protocol uses hop count as its metric?

A. OSPF **B.** RIP **C.** EIGRP **D.** BGP

Answer: B. RIP selects routes using hop count.

Q2. What is RIP's maximum usable hop count?

A. 1 **B.** 15 **C.** 16 **D.** 255

Answer: B. 15 is reachable; 16 represents infinity/unreachable.

Q3. Which protocol uses Dijkstra's shortest path first algorithm?

A. RIP **B.** OSPF **C.** FTP **D.** DHCP

Answer: B. OSPF is a link-state protocol using SPF.

Q4. Which route source has the lowest default Cisco administrative distance?

A. RIP **B.** OSPF **C.** Connected **D.** Static

Answer: C. Connected routes have AD 0.

Q5. What is the purpose of route redistribution?

A. Encrypt routing updates **B.** Exchange routes between different routing protocols **C.** Assign MAC addresses **D.** Create VLANs

Answer: B. Redistribution injects routes from one routing source into another.

16. Backup, Restore and Password Recovery

16.1 Backup startup/running configuration using TFTP

TFTP is a simple file-transfer protocol commonly using UDP port 69. It has minimal features and no built-in authentication/encryption, so use it only in trusted management environments. Ensure IP reachability between the router and TFTP server.

```
copy running-config tftp:
Address or name of remote host? 192.168.1.100
Destination filename? R1-running.cfg

copy startup-config tftp:
copy tftp: startup-config
copy tftp: running-config
```

The exact prompts vary by IOS. Copying to startup-config preserves a backup for next boot; copying to running-config merges/updates the active configuration. Verify files on the TFTP server.

16.2 IOS image backup and restore

Cisco IOS images are commonly stored in flash. Use `show version` and `dir flash:` to inspect the running image and files. Backup/restore may use TFTP or another supported transfer method. Confirm image compatibility, checksum, sufficient flash space and boot variable before replacing an image.

```
show version
dir flash:
copy flash: tftp:
copy tftp: flash:
show boot
```

16.3 Password recovery concepts

Password recovery is performed only on equipment you are authorised to administer. On many Cisco routers, recovery involves interrupting the normal boot process, entering ROMMON, changing the configuration register to ignore startup-config temporarily, booting, merging the saved configuration and setting new credentials, then restoring the normal register and saving. Exact procedure varies significantly by hardware, IOS and security settings.

- Protect physical access to routers; physical console access can enable recovery.
- Do not erase startup-config unless intentionally resetting the device.
- After recovery, restore normal boot settings and verify the device boots with the saved configuration.
- Some devices/platforms have secure boot or password-recovery disable options that change the procedure.

Quick revision and MCQ traps

- TFTP typically uses UDP port 69.
- Running-config is active; startup-config is saved for boot.

- IOS images are commonly stored in flash.
- Password recovery is platform-specific and requires authorised access.

Common traps: TFTP is not encrypted or authenticated by default. • Do not confuse copying a config with copying an IOS image. • A factory reset/erase is not the same as preserving and recovering a configuration.

Q1. Which transport protocol does TFTP use?

A. TCP **B.** UDP **C.** ICMP **D.** SCTP

Answer: B. TFTP uses UDP, with server listening on port 69.

Q2. Where is startup-config commonly stored?

A. RAM **B.** NVRAM **C.** CPU cache **D.** ROM only

Answer: B. Cisco platforms commonly store startup-config in NVRAM.

Q3. Which command lists files in flash?

A. dir flash: **B.** show arp **C.** show vlan **D.** ipconfig

Answer: A. dir flash: lists files in flash storage.

17. Security: Standard and Extended ACL

17.1 What is an ACL?

An Access Control List is an ordered set of permit/deny statements used to classify and control traffic. ACLs are processed top-down; the first matching entry determines the result. An implicit deny exists at the end of a Cisco ACL, so traffic not explicitly permitted is denied.

17.2 Standard ACL

A standard IPv4 ACL filters primarily on source IPv4 address. Numbered standard ACL ranges traditionally include 1–99 and 1300–1999; named standard ACLs use a descriptive name. Because it cannot distinguish destination or transport port, place it close to the destination in many designs.

```
access-list 10 permit 192.168.1.0 0.0.0.255
access-list 10 deny any
interface gigabitEthernet 0/0
ip access-group 10 in
```

17.3 Extended ACL

An extended IPv4 ACL can match source and destination IP addresses, protocol (IP, TCP, UDP, ICMP) and, for TCP/UDP, source/destination ports. Numbered extended ACL ranges traditionally include 100–199 and 2000–2699. Extended ACLs are often placed close to the source to stop unwanted traffic early.

```
access-list 110 permit tcp 192.168.1.0 0.0.0.255 any eq 443
access-list 110 deny ip any any
interface gigabitEthernet 0/1
ip access-group 110 in
```

17.4 Named ACLs, wildcard masks and direction

Named ACLs are created using `ip access-list standard NAME` or `ip access-list extended NAME`. A wildcard mask uses 0 to mean “must match” and 1 to mean “ignore”; it is the inverse of a subnet mask. Example: /24 subnet mask 255.255.255.0 corresponds to wildcard 0.0.0.255. ACL direction is relative to the router interface: **in** filters packets as they enter; **out** filters as they leave.

```
ip access-list extended WEB-ONLY
permit tcp 192.168.10.0 0.0.0.255 any eq 80
permit tcp 192.168.10.0 0.0.0.255 any eq 443
deny ip any any
interface gigabitEthernet 0/0
ip access-group WEB-ONLY in
```

17.5 ACL verification and editing

Command	Purpose
show access-lists	Displays ACL entries and match counters.
show ip interface	Shows ACLs applied to interfaces.
show running-config	Checks ACL definitions and interface bindings.
ip access-group NAME in|out	Applies ACL to an interface.
no ip access-group NAME in|out	Removes ACL from interface.
ip access-list ...	Enters named ACL configuration for editing.

EXAM ALERT: Order matters. Put specific permit/deny statements before broad statements. A broad deny any placed early can make later permit entries unreachable.

Quick revision and MCQ traps

- Standard ACL matches source IP; extended ACL can match source, destination, protocol and ports.
- ACLs are top-down, first-match; implicit deny at end.
- Wildcard 0 = compare; wildcard 1 = ignore.
- IN means entering an interface; OUT means leaving it.
- Standard ACLs are often near destination; extended ACLs often near source.

Common traps: ACL is not processed bottom-up. • A permit entry does not override an earlier deny. • An ACL defined but not applied to an interface has no interface filtering effect.

Q1. Which field is matched by a standard IPv4 ACL?

A. Source IP only **B.** Destination TCP port only **C.** MAC vendor only **D.** DNS hostname

Answer: A. Standard ACLs primarily filter on source IPv4 address.

Q2. What happens if no ACL entry matches a packet?

A. Permit by default **B.** Implicit deny **C.** Send to DNS **D.** Convert to broadcast

Answer: B. Cisco ACLs have an implicit deny at the end.

Q3. What is the wildcard mask for 255.255.255.0?

A. 255.255.255.255 **B.** 0.0.0.255 **C.** 0.0.0.0 **D.** 255.255.0.0

Answer: B. Wildcard is the inverse of the subnet mask.

Q4. Which ACL can filter HTTPS using destination TCP port 443?

A. Standard ACL only **B.** Extended ACL **C.** MAC table **D.** VTP

Answer: B. Extended ACLs can match TCP/UDP ports.

18. Address Translation: NAT, DNAT and PAT

18.1 Why NAT is used

Network Address Translation modifies IP address information in packet headers as traffic passes through a router/firewall. It is widely used to allow private IPv4 networks to communicate with public networks, conserve public IPv4 addresses and publish internal services. NAT is not a security substitute for a firewall.

18.2 NAT terminology

Term	Meaning
Inside local	Address of an inside host as seen on the inside network, often private.
Inside global	Public/global address representing the inside host to outside networks.
Outside global	Real address of an outside host.
Outside local	Address of outside host as represented to the inside network (often same as outside global).

18.3 Static NAT

Static NAT creates a fixed one-to-one mapping between an inside local address and an inside global address. It is useful when an internal server must be reachable using a consistent public address, assuming firewall policy and routing permit it.

```
interface gigabitEthernet 0/0
ip nat inside
interface gigabitEthernet 0/1
ip nat outside
ip nat inside source static 192.168.1.10 203.0.113.10
show ip nat translations
```

18.4 Dynamic NAT

Dynamic NAT maps inside local addresses to an available address from a configured public pool. Mappings are created as needed and are not necessarily permanently tied to a host. If the pool is exhausted, additional hosts cannot obtain a translation until one becomes available.

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat pool PUBLIC 203.0.113.10 203.0.113.20 netmask 255.255.255.0
ip nat inside source list 1 pool PUBLIC
```

18.5 PAT (NAT overload)

Port Address Translation allows many inside hosts to share one public IPv4 address by translating transport-layer port numbers (and associated mapping state). It is also called NAT overload. It conserves public addresses and is the common home/office internet-access pattern.

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet 0/1 overload
```

18.6 DNAT and port forwarding

Destination NAT (DNAT) changes the destination address (and possibly port) of incoming traffic, often publishing an internal service. Cisco IOS static PAT example: `ip nat inside source static tcp 192.168.1.10 80 203.0.113.10 8080` maps traffic arriving at public TCP 8080 to internal TCP 80. The outside interface, routing and firewall/ACL policy must also permit the traffic.

Type	Mapping behaviour	Typical use
Static NAT	Fixed one-to-one address mapping.	Publish a server with a stable public IP.
Dynamic NAT	Temporary one-to-one mapping from a pool.	Translate inside users while pool addresses are available.
PAT / overload	Many-to-one or many-to-few using ports.	Many office clients share one public IP.
DNAT	Changes destination address/port.	Port forwarding to an internal server.

Quick revision and MCQ traps

- NAT changes address information; PAT also uses port numbers.
- Static NAT is fixed one-to-one; dynamic NAT uses a pool; PAT multiplexes many hosts.
- Inside/outside labels are configured on interfaces.

- NAT translations can be checked with show ip nat translations.

Common traps: NAT does not automatically block all unsolicited traffic. • PAT is not the same as static one-to-one NAT. • DNAT is destination translation; SNAT is source translation.

Q1. Which NAT type lets many private hosts share one public IP using ports?

A. Static NAT **B.** Dynamic NAT **C.** PAT **D.** No-NAT

Answer: C. PAT overload uses port numbers to distinguish sessions.

Q2. Which command marks an interface as NAT outside?

A. ip nat outside **B.** ip access-group out **C.** ip route outside **D.** nat enable

Answer: A. The command identifies the outside-facing NAT interface.

Q3. A public TCP port is mapped to an internal server port. This is commonly:

A. DNAT / static PAT **B.** VTP **C.** RIP **D.** ARP

Answer: A. Destination translation/port forwarding maps inbound destination to an internal service.

19. Switching: VLAN, DTP, VTP, STP and Port Security

19.1 VLAN fundamentals

A Virtual LAN is a logical Layer 2 broadcast domain. VLANs segment a switched network, reduce unnecessary broadcast scope and support separation of departments or trust zones. Devices in different VLANs require Layer 3 routing to communicate.

VLAN concept	Meaning
Access port	Normally carries traffic for one data VLAN; frames are usually untagged to the endpoint.
Trunk port	Carries multiple VLANs between network devices using VLAN tags (commonly IEEE 802.1Q).
Native VLAN	VLAN whose traffic is sent untagged on an 802.1Q trunk by default; both ends should match.
Default VLAN	On many Cisco switches, VLAN 1 exists by default; it is not automatically a good choice for all user traffic.
Voice VLAN	Separate VLAN configured on an access port for IP phone traffic (platform-dependent).

19.2 VLAN creation and port assignment

```
enable
configure terminal
vlan 10
name SALES
vlan 20
name HR
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
interface fastEthernet 0/2
switchport mode access
switchport access vlan 20
end
show vlan brief
```

19.3 Trunk configuration

```
configure terminal
interface gigabitEthernet 0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
switchport trunk native vlan 99
end
show interfaces trunk
```

Some switch models require `switchport trunk encapsulation dot1q`; many modern Catalyst models support only 802.1Q and do not expose that command. Ensure VLAN 99 exists and native VLAN settings match at both ends.

19.4 Inter-VLAN routing

Devices in different VLANs are in different IP subnets and need a Layer 3 device to route between them.

Router-on-a-Stick (ROS)

A router uses one physical Ethernet interface configured as a trunk, with one subinterface per VLAN. Each subinterface has an 802.1Q VLAN tag and acts as that VLAN's default gateway.

```
interface gigabitEthernet 0/0
no shutdown
interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

Switch Virtual Interface (SVI)

A multilayer switch can route between VLANs using virtual interfaces. Configure VLANs, assign access ports, create SVIs and enable Layer 3 routing where supported.

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
ip routing
```

19.5 DTP

Dynamic Trunking Protocol is a Cisco proprietary protocol that negotiates trunking between compatible switch ports. Modes include access, trunk, dynamic desirable and dynamic auto (availability varies). For predictable and secure designs, explicitly configure access/trunk mode and disable negotiation where supported.

Mode	Behaviour
access	Forces access mode; does not negotiate trunk.
trunk	Forces trunk mode.
dynamic desirable	Actively attempts to form a trunk.
dynamic auto	Passively becomes trunk if neighbour initiates compatible negotiation.
nonegotiate	Disables DTP negotiation on supported trunk ports.

19.6 VTP

VLAN Trunking Protocol is a Cisco Layer 2 protocol for distributing VLAN database information among switches in a VTP domain. Modes commonly include server, client and transparent; newer versions/platforms may also support off. VTP can simplify administration but a wrong revision/database can propagate unwanted VLAN changes. Verify domain, version, password, mode and revision before connecting switches.

Mode	Typical behaviour
Server	Can create/modify/delete VLANs and advertise changes in a VTP domain.
Client	Receives VLAN database updates; traditionally cannot create/modify VLANs locally.
Transparent	Does not synchronise its local VLAN database like a client; forwards VTP advertisements in supported versions.
Off	Does not participate/forward VTP advertisements (where supported).

19.7 STP

Spanning Tree Protocol prevents Layer 2 switching loops when redundant links exist. A loop can cause broadcast storms, duplicate frames and MAC table instability. STP elects a root bridge, calculates a loop-free logical topology and places redundant ports into a non-forwarding state as appropriate. Classic 802.1D converges relatively slowly; Rapid STP (802.1w) improves convergence. Per-VLAN STP variants run spanning tree per VLAN.

- Root bridge is elected using the lowest Bridge ID (priority plus MAC address; extended system ID contributes VLAN ID in per-VLAN implementations).
- Each non-root switch selects a root port with the best path to the root.
- Each segment has a designated port forwarding toward the root.
- Redundant paths are blocked/discarding to prevent loops, but remain available if topology changes.
- PortFast is for edge ports connected to end devices; combine with BPDU Guard to protect against unexpected switches.

19.8 Port Security

Port security limits which source MAC addresses can use a switch access port. It can use statically configured secure MAC addresses, dynamically learned addresses or sticky learning. A violation occurs when an unauthorised MAC appears or the maximum is exceeded.

Violation mode	Behaviour
protect	Drops violating frames; generally does not generate a syslog/trap or increment violation counter on many IOS implementations.
restrict	Drops violating frames; increments counter and can generate log/trap.
shutdown	Places port into error-disabled state; common default violation action.

```
interface fastEthernet 0/10
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
end
show port-security interface fastEthernet 0/10
```

To recover an error-disabled port, identify and correct the cause, then use `shutdown` followed by `no shutdown` (or configure errdisable recovery where appropriate).

19.9 Switching verification commands

Command	Purpose
show vlan brief	VLAN list and access-port membership.
show interfaces trunk	Trunk status, native VLAN and allowed VLANs.
show spanning-tree	STP root/port roles and states.
show vtp status	VTP domain, mode, version and revision.
show port-security	Port-security summary.
show mac address-table	MAC learning/forwarding table.
show interfaces status	Port state, VLAN and speed/duplex.

Quick revision and MCQ traps

- VLAN creates a separate Layer 2 broadcast domain.
- Trunk carries multiple VLANs; 802.1Q tags identify VLANs.
- Router-on-a-Stick uses router subinterfaces; SVI uses Layer 3 switch virtual interfaces.
- DTP negotiates trunking; VTP distributes VLAN database information.
- STP prevents Layer 2 loops; root bridge is elected using lowest Bridge ID.
- Port security restricts allowed source MAC addresses.

Common traps: A trunk does not itself route between VLANs. • VTP is not the same as DTP: VTP distributes VLAN information; DTP negotiates trunks. • Native VLAN mismatch can cause problems/security exposure. • PortFast is not for switch-to-switch links; BPDU Guard is commonly used on edge ports. • A VLAN ID alone does not create inter-VLAN connectivity.

Q1. Which standard adds VLAN tags to Ethernet frames?

A. 802.1Q **B.** 802.11 **C.** 802.3af **D.** 802.1X only

Answer: A. IEEE 802.1Q is the common VLAN tagging standard.

Q2. Which protocol prevents Layer 2 switching loops?

A. RIP **B.** STP **C.** DHCP **D.** FTP

Answer: B. STP blocks redundant paths to create a loop-free Layer 2 topology.

Q3. What does DTP negotiate?

A. IP addresses **B.** Trunk formation **C.** DNS records **D.** TCP sessions

Answer: B. DTP negotiates trunking between compatible Cisco switch ports.

Q4. Which technology distributes VLAN database information?

A. VTP **B.** STP **C.** PAT **D.** ICMP

Answer: A. VTP distributes VLAN information within a configured domain.

Q5. Which inter-VLAN routing method uses router subinterfaces?

A. Router-on-a-Stick **B.** Port security **C.** VTP transparent **D.** Static NAT

Answer: A. Router-on-a-Stick uses one trunk and tagged subinterfaces.

Q6. Which port-security violation mode commonly disables the port?

A. protect **B.** restrict **C.** shutdown **D.** forward

Answer: C. Shutdown places the interface into an error-disabled state.

20. Practical Lab Checklist and Troubleshooting Method

20.1 Lab-to-syllabus checklist

Lab area	What you should be able to do
Packet Tracer basics	Place devices, select cables, use logical/physical workspace and Simulation mode.
CLI administration	Move between modes; use help; set hostname, banners, passwords; save configuration.
Connectivity	Configure PC/router IPs, masks, gateways; enable interfaces; test with ping.
Layer 2	Inspect MAC table; configure VLANs/access ports/trunks; inspect STP.
Layer 3	Plan subnet schemas; configure physical and loopback interfaces; inspect routing table.
Remote access	Configure SSH; understand why Telnet is insecure.
Routing	Configure static, RIP, EIGRP and single-area OSPF; verify neighbours and routes.
Redistribution	Exchange routes between protocol domains with suitable metrics and filtering.
Backup/restore	Copy configuration and IOS files through TFTP; understand authorised password recovery.
Security	Apply standard/extended ACLs; understand NAC.
NAT	Configure static, dynamic and PAT; understand DNAT/port forwarding.
Switching	Configure VLAN, DTP, VTP, STP, inter-VLAN routing and port security.

20.2 Layer-by-layer troubleshooting

1. **Layer 1:** power, cable type, link light, interface shutdown, speed/duplex and physical errors.
2. **Layer 2:** correct VLAN, access/trunk mode, allowed VLAN list, native VLAN, MAC learning, STP state.
3. **Layer 3:** IP address, subnet mask, default gateway, duplicate IP, route and return route.
4. **Layer 4:** TCP/UDP port, ACL/firewall, listening service and connection state.
5. **Layer 7:** DNS record, application service enabled, correct URL, authentication and server configuration.

20.3 Essential verification commands

Purpose	Useful commands
Device/platform	show version; show running-config; show startup-config
Interfaces	show ip interface brief; show interfaces; show interfaces status
Layer 2	show mac address-table; show vlan brief; show interfaces trunk; show spanning-tree
Layer 3	show ip route; show arp; show ip protocols
Routing neighbours	show ip ospf neighbor; show ip eigrp neighbors
Security/NAT	show access-lists; show ip nat translations; show port-security
Connectivity	ping; traceroute; terminal monitor (where supported)

EXAM ALERT: Troubleshoot from the bottom up. If the physical link is down, changing DNS records will not fix connectivity.

21. High-yield comparison sheet

Compare	Difference to remember
Hub vs Switch	Hub repeats bits to all ports (L1); switch forwards frames by MAC (L2).
Switch vs Router	Switch forwards within VLAN using MAC; router routes between IP networks.
MAC vs IP	MAC is Layer 2/local-link address; IP is Layer 3 logical address.
TCP vs UDP	TCP reliable ordered byte stream; UDP lightweight connectionless datagrams.
HTTP vs HTTPS	HTTPS is HTTP protected by TLS.
POP3 vs IMAP	POP3 retrieves/downloads; IMAP synchronises server mailbox.
SMTP vs POP3	SMTP sends; POP3 retrieves.
DNS vs DHCP	DNS resolves names; DHCP leases network configuration.
FLSM vs VLSM	Same-size subnets vs variable-size subnets.
Static vs Dynamic routing	Manual routes vs protocol-learned routes.
RIP vs OSPF	Distance-vector hop count vs link-state cost/SPF.
Standard vs Extended ACL	Source IP only vs source/destination/protocol/ports.
Static NAT vs PAT	Fixed one-to-one vs many-to-one using ports.
DTP vs VTP	Trunk negotiation vs VLAN database distribution.
STP vs VTP	Loop prevention vs VLAN database distribution.
Router-on-a-Stick vs SVI	Router subinterfaces over trunk vs Layer 3 switch virtual interfaces.
Firewall vs Proxy	Traffic policy enforcement vs intermediary request handling.

22. Final rapid revision: numbers, ports, formulas and commands

22.1 Port numbers to memorise

Port	Protocol/service	Transport
20/21	FTP data/control (traditional active FTP data on 20; control 21)	TCP
22	SSH	TCP
23	Telnet	TCP
25	SMTP relay	TCP
53	DNS	UDP and TCP
67/68	DHCP server/client	UDP
69	TFTP	UDP
80	HTTP	TCP (HTTP/3 uses QUIC/UDP)
110	POP3	TCP
143	IMAP	TCP
443	HTTPS / HTTP over TLS; HTTP/3 uses QUIC	TCP or UDP for QUIC
587	SMTP message submission	TCP
993/995	IMAPS / POP3S	TCP

22.2 Numbers and facts

- OSI = 7 layers; common TCP/IP model = 4 layers (some texts use 5).
- IPv4 = 32 bits; IPv6 = 128 bits; Ethernet MAC = 48 bits.
- TCP minimum header = 20 bytes; UDP header = 8 bytes.
- RIP maximum reachable hop count = 15; 16 = infinity.
- Cisco default AD: connected 0, static 1, EIGRP internal 90, OSPF 110, RIP 120, EIGRP external 170.
- Ethernet broadcast MAC = FF:FF:FF:FF:FF:FF.
- IPv4 private ranges: 10/8, 172.16/12, 192.168/16.
- Loopback IPv4 range = 127/8; common host loopback = 127.0.0.1.
- DHCP DORA = Discover, Offer, Request, Acknowledge.

22.3 Subnetting formula card

IPv4 addresses per subnet = 2^h
Usable hosts (ordinary subnet) = $2^h - 2$
Number of subnets = 2^s
Prefix length = 32 – h
Block size = 256 – mask value in interesting octet

22.4 Cisco command card

enable → configure terminal → hostname R1
interface g0/0 → ip address IP MASK → no shutdown
show ip interface brief → show interfaces
show mac address-table → show vlan brief → show interfaces trunk
ip route NETWORK MASK NEXT-HOP → show ip route
router ospf 1 → network NETWORK WILDCARD area 0
copy running-config startup-config

23. Mixed mock examination — 30 MCQs

Q1. Which device creates separate collision domains for each switch port?

A. Hub **B.** Repeater **C.** Switch **D.** Modem

Answer: C. Each switch port is its own collision domain in normal switched Ethernet.

Q2. Which OSI layer is responsible for end-to-end segmentation?

A. Layer 2 **B.** Layer 3 **C.** Layer 4 **D.** Layer 6

Answer: C. Transport Layer performs segmentation and process delivery.

Q3. Which protocol resolves an IPv4 address to a MAC address on a local link?

A. DNS **B.** ARP **C.** DHCP **D.** FTP

Answer: B. ARP resolves local next-hop IPv4 addresses to Layer 2 MAC addresses.

Q4. Which service leases an IP address automatically?

A. DNS **B.** SMTP **C.** DHCP **D.** IMAP

Answer: C. DHCP dynamically provides client IP configuration.

Q5. Which port is used by SSH by default?

A. 21 **B.** 22 **C.** 23 **D.** 25

Answer: B. SSH commonly uses TCP port 22.

Q6. Which TCP flag is used for graceful connection termination?

A. SYN **B.** FIN **C.** RST **D.** URG

Answer: B. FIN indicates a sender has no more data to send.

Q7. Which protocol is connectionless at the Transport Layer?

A. TCP **B.** UDP **C.** SSH **D.** FTP

Answer: B. UDP sends datagrams without establishing a transport connection.

Q8. Which IPv4 prefix normally provides 62 usable host addresses?

A. /25 **B.** /26 **C.** /27 **D.** /28

Answer: B. /26 leaves 6 host bits: 64 total minus network and broadcast = 62.

Q9. What is the broadcast address for 192.168.10.0/24?

A. 192.168.10.0 **B.** 192.168.10.1 **C.** 192.168.10.254 **D.** 192.168.10.255

Answer: D. All host bits are 1 in the broadcast address.

Q10. Which routing protocol is link-state?

A. RIP **B.** OSPF **C.** EIGRP **D.** ARP

Answer: B. OSPF is link-state and uses SPF.

Q11. Which route is preferred when two routes to the same prefix have different AD values?

A. Higher AD **B.** Lower AD **C.** Higher metric always **D.** Older route

Answer: B. Lower administrative distance is preferred by Cisco routing logic.

Q12. What is the function of a default gateway?

A. Resolve DNS **B.** Forward traffic to other networks **C.** Assign MAC address **D.** Encrypt packets

Answer: B. The host sends off-subnet traffic to its default gateway.

Q13. Which VLAN port type carries multiple VLANs?

A. Access **B.** Trunk **C.** Console **D.** Loopback

Answer: B. Trunk links carry multiple VLANs, commonly with 802.1Q tags.

Q14. Which protocol prevents switching loops?

A. DTP **B.** VTP **C.** STP **D.** RIP

Answer: C. STP blocks redundant Layer 2 paths.

Q15. Which protocol negotiates trunk links on Cisco switches?

A. VTP **B.** DTP **C.** STP **D.** OSPF

Answer: B. DTP negotiates trunking.

Q16. Which protocol distributes VLAN database updates?

A. DTP **B.** VTP **C.** DHCP **D.** DNS

Answer: B. VTP distributes VLAN information within a domain.

Q17. Which NAT method uses port numbers to share one public address?

A. Static NAT **B.** Dynamic NAT **C.** PAT **D.** DNAT only

Answer: C. PAT overload uses ports to distinguish flows.

Q18. Which ACL type can match destination TCP port 443?

A. Standard **B.** Extended **C.** MAC-only **D.** VTP

Answer: B. Extended ACLs match protocol and TCP/UDP ports.

Q19. What does the implicit deny mean?

A. Unmatched traffic is permitted **B.** Unmatched traffic is denied **C.** Only UDP is denied **D.** Only broadcasts are denied

Answer: B. Cisco ACLs deny packets that reach the end without a match.

Q20. Which command saves running configuration?

A. write erase **B.** copy running-config startup-config **C.** reload **D.** show startup-config

Answer: B. It writes the active configuration to startup storage.

Q21. Which storage is commonly used for Cisco IOS images?

A. Flash **B.** CPU register **C.** Cache only **D.** Console line

Answer: A. IOS images are commonly stored in flash.

Q22. Which protocol is used for simple file transfer and commonly uses UDP 69?

A. FTP **B.** TFTP **C.** SFTP **D.** HTTP

Answer: B. TFTP commonly uses UDP port 69.

Q23. Which is the correct DHCP sequence?

A. DORA **B.** RADO **C.** ODRA **D.** ADRO

Answer: A. Discover, Offer, Request, Acknowledge.

Q24. Which DNS record identifies mail exchangers?

A. A **B.** PTR **C.** MX **D.** CNAME

Answer: C. MX records identify mail servers for a domain.

Q25. Which topology has a central connecting device?

A. Bus **B.** Star **C.** Ring **D.** Full mesh

Answer: B. Star topology connects endpoints to a central hub/switch.

Q26. Which feature distributes voice/video traffic preferentially?

A. QoS **B.** NAT **C.** ARP **D.** VTP

Answer: A. QoS prioritises traffic according to policy.

Q27. What is the main function of a reverse proxy?

A. Represent backend servers **B.** Assign host IPs **C.** Prevent all Layer 2 loops **D.** Replace DNS

Answer: A. A reverse proxy sits in front of servers and handles client requests.

Q28. Which IPv6 statement is true?

A. IPv6 is 64 bits **B.** IPv6 uses broadcast **C.** IPv6 is 128 bits **D.** IPv6 uses dotted decimal

Answer: C. IPv6 addresses are 128 bits.

Q29. Which command shows learned switch MAC addresses?

A. show ip route **B.** show mac address-table **C.** show ip protocols **D.** show version only

Answer: B. This command displays the Layer 2 forwarding database.

Q30. Which technology controls which devices are admitted to a network?

A. NAC **B.** PAT **C.** DNS **D.** STP

Answer: A. NAC evaluates identity/device posture and enforces access policy.

24. Final exam checklist

- I can name all seven OSI layers in both directions and state each PDU.
- I can map OSI to TCP/IP and explain encapsulation/de-encapsulation.
- I know common devices, network types, topologies and unicast/broadcast/multicast.
- I know the listed application protocols, their purposes and default ports.
- I can compare TCP, UDP and ICMP and explain the TCP handshake.
- I can calculate network, broadcast, usable range, host count and prefix for IPv4 subnets.
- I can distinguish FLSM and VLSM and allocate variable-size blocks.
- I can navigate Cisco CLI modes, configure hostname/password/banner and save configuration.
- I can configure IP interfaces, loopbacks, SSH, static routes, RIP, EIGRP and OSPF.
- I understand redistribution, TFTP backup/restore, ACLs, NAT/PAT, VLANs, DTP, VTP, STP and port security.
- I can troubleshoot bottom-up and use the relevant show commands.

End of study notes

These notes cover the theory and practical topics visible in the two uploaded syllabus images. Use Packet Tracer to practise the configurations: command syntax and feature support can differ by device model and IOS image. For a production network, follow the organisation's security and change-management standards.